DEPARTMENT OF PHYSICS
KYOTO UNIVERSITY.

集中講義
柳田

No. 1

26 (水) Gauge Symmetry 自発的対称性の破れ (Nanbu ... '61)
27 (木) ⌟ Chiral Theory ($\pi^{\pm}, \pi^{0}$) Gleshow
28 (金) 66 JJ. Sakurai ρ-meson → Anderson '61 Higgs mechanism → Weinberg-Salam Electo-weak 統一理論

/

3 (水) Electroweak 理論 バリオン数非保存
4 (木)
5 (金) 大統一理論

I Gauge Fields and Symmetries

$$S = \int dt\, L(t) = \int d^4x\, \mathcal{L}(\varphi_\alpha(x), \partial_\mu \varphi_\alpha(x)) \qquad \varphi_\alpha(x) = \int \frac{d^3\vec{k}}{(2\pi)^3 2\omega_k} \left(a(\vec{k}) e^{-ikx} + b^+(\vec{k}) e^{ikx} \right)$$

(1) Variational Principle

粒子 $a^+(k)|0\rangle = |k = \vec{k}, k_0\rangle$

反粒子 $b^+(k)|0\rangle = |k\rangle$

$$\delta^* \int \mathcal{L}\, d^4x = 0 \rightarrow \frac{\partial \mathcal{L}}{\partial \varphi_\alpha} - \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_\alpha)} \right] = 0$$

spin 0

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \varphi)^2 + \frac{1}{2} m^2 \varphi^2 \qquad m^2\varphi - \partial_\mu[-\partial_\mu \varphi] = 0 \quad \text{Klein-Gordon eq}$$

(2) Symmetry

Example. U(1) Complex field $\varphi(x) \longmapsto e^{-i\theta g}\varphi(x)$

$\varphi^*(x) \longmapsto e^{i\theta g}\varphi^*(x)$ 対称性

$\varphi \rightarrow \varphi' = \varphi + \delta\varphi$

$\boxed{\delta \mathcal{L} = 0}$ $\qquad \delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \varphi}\delta\varphi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)}\delta(\partial_\mu\varphi) = \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)}\right]\delta\varphi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)}\partial_\mu\delta\varphi = \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)}\delta\varphi\right] = 0$

$\frac{\partial \mathcal{L}}{\partial(\partial_\mu\varphi)} = J_\mu$

$\partial_\mu J_\mu = 0 \qquad Q(t) = \int J_0\, d^3x \qquad \frac{dQ}{dt} = \int \frac{dJ_0}{dt} d^3x = \int \partial_i J_i\, d^3x = 0$

保存する

$\dot{Q} = [\hat{Q}, H] = 0$

粒子 $|\vec{p}, E, m, Q\rangle$

無限小変換 $\varphi \rightarrow e^{-i\epsilon g}\varphi \simeq 1 - i\epsilon g \varphi$

$\therefore \delta\varphi = -i\epsilon g \varphi$

変換パラメータ Charge

$[\varphi, \hat{Q}] = g\varphi$

$J_\mu(x) = -\partial_\mu\varphi^*(-ig\varphi) - \partial_\mu\varphi(ig\varphi)$

$= ig(\partial_\mu\varphi^*\varphi - \varphi^*\partial_\mu\varphi)$ $\qquad Q = \int d^3x\, J_0(x) = i\int d^3x\,(\varphi^*\dot{\varphi} - \varphi\dot{\varphi}^*)$

$\varphi \to U\varphi U^\dagger = e^{-i\varepsilon Q}\varphi\, e^{i\varepsilon Q} \simeq \varphi - i\varepsilon[\hat{Q},\varphi] + \cdots$

$\qquad\qquad\qquad

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \longrightarrow \partial_\mu(A_\nu - \frac{1}{e}\partial_\nu\theta) - \partial_\nu(A_\mu - \frac{1}{e}\partial_\mu\theta) = F_{\mu\nu}$

$(F_{\mu\nu})^2 \Rightarrow (E, H)$ $F_{0i} = E_i$ 54 Yang-Mills 場

$F_{ij} = \epsilon_{ijk} H_k$

$(F_{\mu\nu})^n$ ← くり込み可能性を破る $\tilde{F}_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma} F_{\rho\sigma}$ $\mathcal{L} = F_{\mu\nu}\tilde{F}_{\mu\nu} = \partial_\mu K_\mu \neq 0$

ゲージ場の質量項 ゲージ粒子は massless

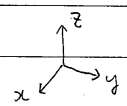
$\mathcal{L}_{mass} = m^2 A_\mu^2 \longrightarrow m^2$ If ゲージ変換のもとに $\varphi \to e^{-i\theta g}\varphi$

$A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\theta$ But $\boxed{\varphi \to \varphi - \theta}$ $\delta\varphi = g\varphi$

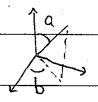
$\delta\varphi = \theta$ Non-Linear 表現

↑ const

O(3)



$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = O \begin{pmatrix} x \\ y \\ z \end{pmatrix}$



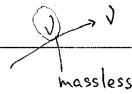
$\begin{cases} \theta \to \theta + a \\ \varphi \to \varphi + b \end{cases}$

◀

$"A_\mu" = A_\mu + \frac{1}{e}\partial_\mu\varphi$ $"A_\mu" \longrightarrow A_\mu - \frac{1}{e}\partial_\mu\theta - \frac{1}{e}\partial_\mu\varphi + \frac{1}{e}\partial_\mu\theta = "A_\mu"$

$\mathcal{L} = m_A^2 ("A_\mu")^2$ はゲージ不変である φ 物理的になりうるか？

$\varphi = \begin{pmatrix} e \\ \nu \end{pmatrix} \to e^{i\vec{\theta}(x)\vec{\tau}} \begin{pmatrix} e \\ \nu \end{pmatrix}$



$\langle\varphi\rangle$ = condensation

Nambu 理論 $p, n \sim 1\,GeV$

$\pi^\pm \pi^0 \sim 0.135 \sim 0.14\,GeV$

Non-Linear 表現

$\begin{pmatrix} \pi^\pm \\ \pi^0 \end{pmatrix} \to$ ~~Constant + ···~~ $m_\pi^2 \pi^2$ は禁止 $\to m_\pi^2 c + \cdots$

U(1) global $\varphi \to e^{i\theta}\varphi$

$\mathcal{L} = \partial_\mu\varphi^*\partial_\mu\varphi + m^2\varphi^*\varphi + \lambda(\varphi^*\varphi)^2$

$\langle\varphi\rangle$ = constant = v φ real 2

$\varphi = (v + \eta(x))e^{i\chi(x)}$ $\langle\chi\rangle = \langle\eta\rangle = 0$

$\mathcal{L} = -\partial_\mu((v+\eta(x))e^{-i\chi(x)})\partial_\mu((v+\eta(x))e^{i\chi(x)}) + m^2(v+\eta)^2$

$= -\partial_\mu\eta\,\partial_\mu\eta + v^2\partial_\mu\chi\,\partial_\mu\chi + \eta^2(\partial_\mu\chi)^2$ $+\lambda(v+\eta)^4$

$+ \cdots + V(\eta, v)$ $V(\varphi^*\varphi)$

$\chi \to \chi + \theta$ $"V(\eta, v)"$

II Dynamical Symmetry Breaking 1961

簡単な例

フェルミオン $\frac{1 \pm \gamma_5}{2} = \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & 0 & \\ & & & 1 \end{pmatrix}, \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$ $\frac{1+\gamma_5}{2}\begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ 0 \\ 0 \end{pmatrix} = \psi_R$ $\frac{1-\gamma_5}{2}\begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \cdot \\ \cdot \end{pmatrix} = \psi_L$

$[\Sigma_{\mu\nu}, \gamma_5] = 0$ ワイル フェルミオン

massless fermion ψ_L $\longrightarrow$ p

if massive $v<c$ $\rightleftharpoons \psi_R$ ψ_L, ψ_R わけられない $\not{\partial} = \partial_\mu \gamma_\mu$

$$\mathcal{L} = \bar{\psi}_L \not{\partial} \psi_L + \bar{\psi}_R \not{\partial} \psi_R + \underset{\parallel}{m} \psi\psi + \underset{\parallel}{m} \bar{\psi}\bar{\psi}$$
$$0 \qquad 0$$

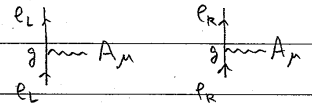
$\begin{cases} \psi_L \to e^{i\alpha} \psi_L \\ \psi_R \to e^{i\beta} \psi_R \end{cases}$ global $U(1)_L \times U(1)_R$

$U(1)_L \times U(1)_R = U(1)_A \times U(1)_V$ $\psi = \psi_L + \psi_R$

$U(1)_V : U(1)_{L+R}$ $\begin{cases} \psi_L \to e^{i\alpha} \psi_L \\ \psi_R \to e^{i\alpha} \psi_R \end{cases}$ $\psi \to e^{i\alpha} \psi$ $m\bar{\psi}\psi$: $U(1)_V$ で invariant

$U(1)_A : U(1)_{R-L}$ $\begin{cases} \psi_L \to e^{-i\beta} \psi_L \\ \psi_R \to e^{+i\beta} \psi_R \end{cases}$ $\psi \to e^{i\alpha\gamma_5} \psi$ $U(1)_A$ では不変でない

$U(1)_V \times U(1)_A$ ゲージ化する

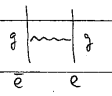


$\mathcal{L} = \bar{e}_L \not{\partial} e_L + \bar{e}_R \not{\partial} e_R$ $D_\mu = \partial_\mu - igA_\mu$

~~$m e_R e_L$~~

$\mathcal{L} = \bar{\psi}_L \not{\partial} \psi_L + \bar{\psi}_R \not{\partial} \psi_R \Rightarrow \bar{\psi}_L \gamma_\mu \partial_\mu \alpha \psi_L e^\alpha + \bar{\psi} \gamma_\mu \partial_\mu \psi_L e^\alpha + \bar{\psi}_R \gamma_\mu \partial_\mu \alpha \psi_R + \bar{\psi}_R \gamma_\mu \partial_\mu \psi_R$

$\bar{\psi}_L \gamma_\mu (\partial_\mu - ig \underset{\parallel}{A_\mu}) \psi_L + \bar{\psi}_R \gamma_\mu (\partial_\mu - igA_\mu) \psi_R$
$\partial\alpha$



$\alpha_\infty = \frac{1}{137}$

$e(x) = e_L(x) + e_R(x)$

A Simple Ex QED (A_μ, e_L, e_R) $\alpha_{em} = \frac{g^2}{4\pi} = \frac{1}{137}$ $\alpha \sim O(1)$

$\mathcal{L} = \bar{e}_L \not{\partial} e_L + \bar{e}_R \not{\partial} e_R - \frac{1}{4g^2} F_{\mu\nu} F_{\mu\nu}$

$D_\mu = \partial_\mu - igA_\mu$ 強い coupling をもつ理論 $\langle \bar{e}e \rangle \neq 0$

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ e は massless

Symmetry $U(1)_V \times U(1)_A$ ← global Symmetry
↑ local gauge symmetry

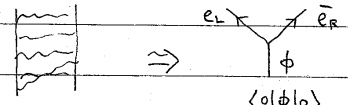
$U(1)_V$ $e_{L,R} \to e^{i\alpha(x)} e_{L,R}$ massterm

$U(1)_A$ $\begin{cases} e_L \to e^{i\beta} e_L \\ e_R \to e^{-i\beta} e_R \end{cases}$ $\mathcal{L}_{mass} = m \bar{e}_R e_L + h.c.$

$\downarrow$

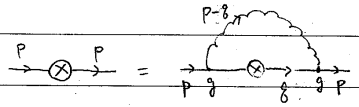
$m e^{i2\beta} \bar{e}_R e_L$

◎ mass が dynamical に generate されるか？



$\langle \bar{e}_R \quad e_L \rangle \rightsquigarrow \bar{e}_R e_L \langle \phi \rangle$
$\|$
0

Schwinger - Dyson 方程式で $e = e_L + e_R$

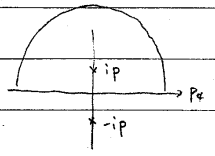


$m = \Sigma(0)$

$$\Sigma(p) = \frac{g^2}{4\pi^2}\int d^4q \, \frac{1}{(p-q)^2}\,\frac{\Sigma(q)}{q^2+\Sigma^2(q)}$$

$$(2\pi)^2\varphi(x) = \int \frac{1}{p^2} e^{ipx} d^4p = \int \frac{e^{i\vec{p}\vec{x}} e^{ip_4x_4}}{\vec{p}^2+p_4^2} d^3\vec{p}\, dp_4$$

$$\int \frac{e^{ip_4x_4}}{\vec{p}^2+p_4^2} dp_4 = \int\left\{\frac{1}{p_4-ip} - \frac{1}{p_4+ip}\right\}\frac{1}{2ip} e^{-ip_4x_4} dp_4$$

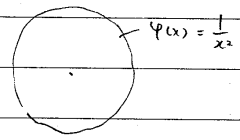


$x_4 > 0 \quad = \frac{2\pi i}{2ip} e^{-px_4}$

$$\int p^2 dp\, d\cos\theta\, e^{ipx\cos\theta}\, 2\pi \frac{\pi}{p} e^{-px_4}$$

$$= 2\pi^2 \int p\,dp \left.\frac{e^{ipx\cos\theta}}{ipx}\right|_{-1}^{1} e^{-px_4} = 2\pi^2 \int p\,dp\, e^{-px_4}\left(\frac{e^{ipx}}{ipx} - \frac{e^{-ipx}}{ipx}\right) = \frac{2\pi^2}{ix}\int dp\left\{e^{-p(x_4-ix)} - e^{-p(x_4+ix)}\right\}$$

$$= \frac{2\pi^2}{ix}\left[\frac{1}{x_4-ix} - \frac{1}{x_4+ix}\right] = \frac{2\pi^2}{ix}\cdot\frac{2ix}{x_4^2+x^2} = \frac{(2\pi)^2}{x^2}$$



$\Box \frac{1}{x^2} = -(2\pi)^2\delta^4(x)$

x: Euclidian $(x_1\ x_2\ x_3\ x_4)$

massless scalar 場で $\Box\varphi(x) = \delta(x)$ ← point source

$$\varphi(x) = \int a_p e^{ipx}\frac{d^4p}{(2\pi)^4}$$

$$-\int p^2 a_p e^{ipx}\frac{d^4p}{(2\pi)^4} = -(2\pi)^2\int \frac{d^4p}{(2\pi)^4} e^{ipx}$$

$$\downarrow$$

$$a_p = \frac{(2\pi)^2}{p^2}$$

$$\varphi(x) = \int \frac{1}{p^2} e^{ipx}\frac{d^4p}{(2\pi)^4} = \frac{1}{x^2}$$

$$\Sigma(p) = \frac{g^2}{4\pi^2}\int d^4q\,\frac{1}{(p-q)^2}\,\frac{\Sigma(q)}{q^2+\Sigma^2(q)}$$

$$\Box_p = \frac{\partial^2}{\partial p_1^2} + \frac{\partial^2}{\partial p_2^2} + \frac{\partial^2}{\partial p_3^2} + \frac{\partial^2}{\partial p_4^2} = \partial_p^2 + \frac{3}{p}\partial_p$$

$$\left(\partial_p^2 + \frac{3}{p}\partial_p\right)\Sigma(p) = -g^2\frac{\Sigma(p)}{p^2+\Sigma^2(p)}$$

$$\Box_p\Sigma(p) = \frac{g^2}{4\pi^2}\int d^4q\,\Box_p\frac{1}{(p-q)^2}\,\frac{\Sigma(q)}{q^2+\Sigma^2(q)} = -g^2\frac{\Sigma(p)}{p^2+\Sigma^2(p)}$$

Def $t \equiv \log p^2/M^2 = 2\log\frac{p}{M}$ $\quad \partial t = 2\frac{1}{p}\partial p$ $\quad \frac{\partial\Sigma(p)}{\partial p} = e^{-\omega(t)} - p\frac{\partial}{\partial p}\left[e^{-\omega(t)}\right] = e^{-\omega(t)} + 2\frac{\partial}{\partial t}\left[e^{-\omega(t)}\right]$

$\Sigma(p) \equiv p\exp[-\omega(t)]$ $\quad p\frac{\partial}{\partial p} = 2\frac{\partial}{\partial t}$ $\quad = e^{-\omega(t)} - 2\omega' e^{-\omega(t)} = e^{-\omega(t)}[1-2\omega']$

$F(t) \equiv \frac{d\omega}{dt}$ $\quad p\frac{\partial^2\Sigma}{\partial p^2} = p\frac{\partial}{\partial p}\left(\frac{\partial\Sigma}{\partial p}\right) = 2\frac{\partial}{\partial t}\left[e^{-\omega}(1-2\omega')\right]$

$= 2\left[-\omega' e^{-\omega}(1-2\omega') + e^{-\omega}(-2\omega'')\right] = e^{-\omega}\left[-2\omega' + 4\omega'^2 - 4\omega''\right]$

$$\left[-2\omega' + 4\omega'^2 - 4\omega''\right] + 3\left[1 - 2\omega\right] = -g^2 \frac{P^2}{P^2 + P^2 e^{-2\omega}} = -g^2 \frac{1}{1+e^{-2\omega}}$$

$$4\omega'' = 4(1-\omega')^2 - 1 + g^2 (1+e^{-2\omega})^{-1}$$

$$\rightarrow \quad 4F' = 4(1-F)^2 - 1 + g^2 (1+e^{-2\omega})^{-1}$$

Boundary Condition

(1) $P^2 \to 0 \quad t \to -\infty \quad \omega \to -\infty$

$F \to \frac{1}{2} \qquad \omega = \frac{1}{2} t$

$$\sum(p) = P e^{\ln(P/M)^{-1}} = P \cdot \frac{M}{P} \simeq \text{const}$$

$m = \sum(0) = \text{const} \neq 0$

$P e^{-\omega} \to \text{const}$

$\omega \sim \ln P \sim t$

$F = \frac{d\omega}{dt}, \quad F' = 0$

$\sum(p) \xrightarrow{P\to\infty} 0$

(2) $P^2 \to \infty \quad t \to \infty \quad \sum(p) \to 0$

$$F' \doteq (1-F)^2 - \frac{1}{4}(1-g^2)$$

$$\frac{d\omega}{dt} = F = 1 \pm \frac{1}{2}\sqrt{1-g^2} \equiv \gamma$$

$\omega = \gamma t$

$$\sum(p) = P e^{-\gamma \ln P^2/M^2} = P^{1-2\gamma}$$

$P e^{-\omega} \sim \frac{1}{P^n}$

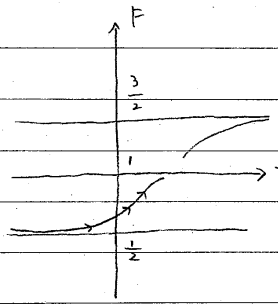
$e^{\omega} \sim P^{n+1}$

$\omega \sim (n+1)\ln P$

◀ $g^2 \to 0$

γ { $1 - \frac{1}{2}$: $\sum(p) = \text{const}$ ← 手で m を加えた explisit breaking
$1 + \frac{1}{2}$: $\sum(p) = \frac{1}{P^2} \to 0$

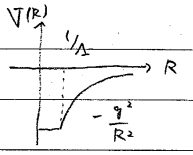
$F \xrightarrow[t\to-\infty]{} \frac{1}{2} \xrightarrow[t\to+\infty]{} 1 + \frac{1}{2}\sqrt{1-g^2}$



(1) $g^2 < 1$

(2) $g^2 > 1$

$g^2 > 1$



* Linearized S.D. 方程式

$$\sum(p) = \frac{g^2}{4\pi^2} \int d^4q \, \frac{1}{(p-q)^2} \frac{\sum(q)}{q^2 + \underbrace{\sum^2(q)}_{m^2}}$$

$$\psi(p) \equiv \frac{\sum(p)}{p^2+m^2} \qquad (p^2+m^2)\psi(p) = \frac{g^2}{4\pi^2} \int d^4q \, \frac{1}{(p-q)^2} \psi(q) \quad \rightleftarrows \quad \left\{ \frac{\partial^2}{\partial R^2} + \frac{3}{R}\frac{\partial}{\partial R} - m^2 - V(R) \right\} \psi(R) = 0 \qquad V(R) = -\frac{g^2}{R^2}$$

$$\psi(R) \equiv \int d^4p \, e^{ipR} \psi(p)$$

$$V(R) = \begin{cases} -\frac{g^2}{R^2} & R \geq \frac{1}{\Lambda} \\ -g^2\Lambda^2 & 0 < R < \frac{1}{\Lambda} \end{cases} \qquad \psi(R) = \begin{cases} A \frac{K_0(R)}{R} & (R > \frac{1}{\Lambda}) \\ B \frac{J_1(\beta)}{\beta} & (0 \leq R < \frac{1}{\Lambda}) \end{cases}$$

$\psi(R) \xrightarrow[R\to 0]{} \text{const}$

$\psi(p) \sim (p^2)^{-\frac{1}{2}(3+s)} = (p^2)^{-2}$

$\langle \bar{e} e \rangle \neq 0$

$$\psi(x) = \int d^4p \, e^{ipx} \psi(R) = \frac{4\pi^2}{x} \int_0^\infty dp \, p^2 J_1(px) \psi(p$$

$$= \frac{4\pi^2}{x^{1-s}} \int_0^\infty dy \, f(y) \qquad s = 1$$

対称性の破れ

$U(1)_A \quad \underbrace{\bar{e}_R e_L}_{\phi} \to e^{i2\beta} \bar{e}_R e_L$

if $\langle e_R e_L \rangle = \text{const}$ $\qquad \psi(p) \sim p^{-4}$

$\langle\phi\rangle \neq 0$ が起こるポテンシャル

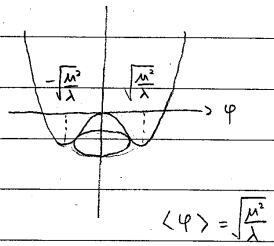
$\mathcal{L} = \partial_\mu \varphi^* \partial_\mu \varphi - V(\varphi)$ $\quad \partial_\mu \partial_\mu \varphi - \frac{\partial V}{\partial \varphi^*} = 0$ $\quad \varphi = 0 \quad |\varphi|^2 = \frac{\mu^2}{\lambda} \neq 0$

$V(\varphi) = -\mu^2 |\varphi|^2 + \frac{\lambda}{2} |\varphi|^2$ $\quad \Box\varphi = -\mu^2 \varphi + \lambda |\varphi|^2 \varphi$

$= (-\mu^2 + \lambda |\varphi|^2)\varphi$

$U(1)_A$ のもとに

$\phi(x) \to e^{i\beta} \phi(x)$



$\langle\varphi\rangle = \sqrt{\frac{\mu^2}{\lambda}}$

54 Yang Mills

61 Nambu - Goldstone ($\pi^\pm \pi^0$)

60 J.J. Sakurai ρ がゲージ粒子

63 Anderson

64 Higgs

67 Weinberg - Salam

III Low-Energy Theory of QCD — Chiral Theory —

Quark $q_{R,L} = \begin{pmatrix} u \\ d \end{pmatrix}_{R,L}$

$\langle \bar{q} q \rangle = v \neq 0$

Global Symmetry $SU(2)_L \times SU(2)_R$ $\quad SU(2)_L \times SU(2)_R \to SU(2)_V$

$q_L \to e^{i\theta_i \sigma_i} q_L$ $\quad (\theta_i, \xi_i) \neq 6$ real parameters $\quad \langle \bar{q}_R q_L \rangle = SU(2)_V$ 不変

$q_R \to e^{i\xi_i \sigma_i} q_R$ $\quad SU(2)_V$ $\{ q_{L,R} \to e^{i\theta_i \sigma_i} q_{L,R}$

massless pion が現れる

π_i $(i = 1, 2, 3)$

$U(1)$ $\phi \to e^{i\theta} \phi$

$\langle \Phi \rangle = v$ $\quad \phi = (v + \eta) e^{i\varphi}$

$U(1)$ $\varphi \to \varphi + \theta$

(1) Non-Linear realization of the Symmetry

群 $G \to H$

$\wedge$ $\quad \uparrow$

$SU(2)_L \times SU(2)_R$ $\quad SU(2)_V$

$(t^\alpha)_{\beta\gamma} = i f_{\alpha\beta\gamma}$

Lie 代数 $T_\alpha = T_a + T_i$ $\quad G$ の Lie 代数 $[T_\alpha, T_\beta] = i f_{\alpha\beta\gamma} T_\gamma$ $\quad [T_\alpha, \varphi_i] = i A_{\alpha i}(\varphi)$

$\uparrow H$ $\quad G/H$ $\quad$ N-G boson φ_i $\quad$ Jacobi-identity

$[T_\alpha [T_\beta \varphi_k]] + [T_\beta [\varphi_\alpha, T_k]] + [\varphi_k, [T_\alpha, T_\beta]] = 0$

A solution

$A_{\alpha i}(\varphi) = \{ e^{i\varphi_k t^k} \}_{\alpha\beta} (e^{-1})_{\beta i} = 1 + \varphi + \varphi^2 + \cdots$ $\quad A_{\alpha i}(\varphi)$ が決まる

$(e^{-1})_{ki} (e)_{il} = \delta_{kl}$ $\Leftarrow$

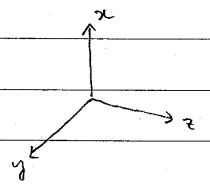
$(e)_{il} = \left[\frac{1 - e^{i\varphi_k t^k}}{-i\varphi_k t^k} \right]_{il}$

$\varphi_i \xrightarrow{G} \varphi_i' = \varphi_i + \varepsilon_i A_{ij}(\varphi)$

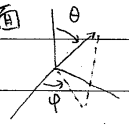
$\mathcal{L} = \partial_\mu \varphi_i \, g_{ij}(\varphi) \, \partial_\mu \varphi_j$

G non-Linear 表現

O(3) $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to O \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $x^2+y^2+z^2=1$



球の表面 $\pi = (\theta, \varphi)$



$\begin{cases} \theta \to \theta + c \\ \varphi \to \varphi + c' \end{cases}$

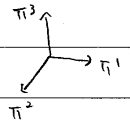
Quark $\begin{pmatrix} u \\ d \end{pmatrix}_L \begin{pmatrix} u \\ d \end{pmatrix}_R$

Chiral $SU(2)_L \times SU(2)_R$

$\langle Q\bar{Q} \rangle \neq 0$ $\hookrightarrow SU(2)_V$

$T^i_{L,R}$ $(i=1\sim3)$ $T^i_{L-R} \Rightarrow \pi^i = (\pi^\pm, \pi^0)$

G/H 座標



$\mathcal{L} = \mathcal{L}_{eff}(\partial_\mu \pi^i)$

↑

G 変換不変

$\mathcal{L}_{eff}$ の形が決定される

linear 表現 $[T^i_\alpha, \varphi_j] = i f_{ijk} \varphi_k$

$[T^i_\alpha, \varphi_j] = i A_{ij}(\varphi)$

Jacobi Id $[T_\alpha [T_\beta, \varphi_i]] + [T_\beta [\varphi_i, T_\alpha]] + [\varphi_i [T_\alpha T_\beta]] = 0$

G 変換 $\varphi_i \to \varphi_i + i \varepsilon_j A_{ij}(\varphi)$ $\mathcal{L} = d\varphi_i \, g_{ij}(\varphi) \, d\varphi_j$

(↑ $x_\mu \to x_\mu + i \varepsilon_\nu f_{\mu\nu}(x)$)

例 $SO(3)/SO(2) = SU(2)/U(1) = G/H$ ～ Köhler manifold (complex)

$SO(3)$ T_1 T_2 T_3 $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$[T_i, T_j] = i \epsilon_{ijk} T_k$ $T_i = \frac{1}{2} \sigma_i$

Unbroken T_3

Broken $T_{1,2}$ ～ N.G modes φ_1, φ_2 (↑ ↑ real)

$\begin{cases} T_+ = T_1 + i T_2 & [T_+, T_-] = 2 T_0 \\ T_- = T_1 - i T_2 & [T_0, T_+] = T_+ \\ T_0 = T_3 & [T_0, T_-] = -T_- \end{cases}$

Broken generator

$T_+ (T_+^*)$ → Nambu-Goldstone

T_- 座標 φ_+ (complex)

$\begin{cases} [T_0, \varphi_+] = \varphi_+ \\ [T_+, \varphi_+] = i \varphi_+^2 \\ [T_-, \varphi_+] = i \end{cases}$ $[T_+ [T_- \varphi_+]] + [T_- [\varphi_+, T_+]] + [\varphi_+ [T_+ T_-]] = 0$

$(T_+, i) + [T_-, -i\varphi_+^2] + [\varphi_+, 2T_0] = 0$

$2\varphi_+ + 2(-\varphi_+) = 0$

$\mathcal{L} = \partial_\mu \varphi^* \frac{1}{(1+\varphi\varphi^*)^2} \partial_\mu \varphi$

$\begin{cases} [T_0, \varphi_+^*] = -\varphi_+^* \\ [T_+, \varphi_+^*] = i \varphi_+^{*2} \\ [T_-, \varphi_+^*] = i \end{cases}$

$[T_0 \mathcal{L}] = 0$ $[T_0, \varphi^* \varphi] = -\varphi^* \varphi + \varphi^* \varphi = 0$ $[T_+ \varphi^* \varphi] = i\varphi + \varphi^* \cdot i\varphi^2 = i\varphi(1+\varphi^*\varphi)$

$[T_+ \mathcal{L}] = \left[T_+, \frac{1}{(1+\varphi\varphi^*)^2} \partial_\mu \varphi^* \partial \varphi\right] = -2 \frac{i(1+\varphi^*\varphi)\varphi}{(1+\varphi^*\varphi)^2} \partial_\mu \varphi^* \partial_\mu \varphi + \frac{1}{(1+\varphi^*\varphi)^2} \partial_\mu \varphi^* \cdot i \partial_\mu \varphi^2 = -2i \frac{\varphi}{(1+\varphi^*\varphi)^2} \partial_\mu \varphi^* \partial_\mu \varphi + 2i \frac{\varphi}{(1+\varphi^*\varphi)^2} \partial_\mu \varphi^* \partial_\mu \varphi$

($\partial_\mu \varphi^2 = 2\varphi \partial_\mu \varphi$)

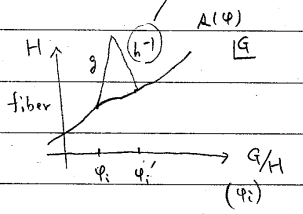
$= 0$

Colman - Wess - Zumino Phys. Rev. 177 (1969) 2239

Define $A(\varphi) = e^{i\varphi_i T_i} \in G$ (↑ G/H)

Broken generator T_i $A: G/H \to G$ mapping

"gauge 変換"



(2) Hidden local symmetry による chiral theory or construction

$G \longrightarrow H \{G/H\}$

linear 表現 φ_α φ_i (Nambu-Goldstone 座標)

自由度 > 自由度

(x,y,z) (θ,φ)

$g \in G$ $A(\varphi) \to A(\varphi') = g A(\varphi) h^{-1}(\varphi, g)$

$G \times H_{local}$ Theory

自由度 $\varphi_\alpha - H_{local}$ = physical な自由度

linear 表現

$\varphi(x) = \eta(x) e^{i\chi(x)} \longrightarrow \eta(x) e^{i\chi(x) + i\theta(x)} = \eta(x)$

gauge 変換

例 CP^N model $= U(N+1)/U(N)\times U(1)$

$A_{ij} =$ ($U(N)$ | $N+1$; $U(1)$) ↑ $N+1$ ↓

← $N+1$ → X_i X_i^*

Nambu - Goldstone

z_i $(i = 1 \sim N)$ complex

$Z = \begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{pmatrix} \longrightarrow e^{i\theta_i \lambda^i} \begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_n \end{pmatrix}$ G

$\mathcal{L} = \partial_\mu Z^* \partial_\mu Z$

$G \times H_{local}$

$Z \to e^{i\alpha(x)} Z$

$\mathcal{L} = D_\mu Z^* \partial_\mu Z = \{(\partial_\mu + iA_\mu) Z^*\}\{(\partial_\mu - iA_\mu) Z\}$

自由度 Z : $2(N+1)$

Gauge の自由度 1

$2N+1$

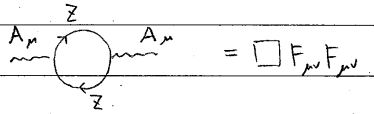
Constraint $Z^* Z = 1$ (2N)

$\mathcal{L} = A_\mu^2 Z^* Z + iA_\mu (Z^* \partial_\mu Z - \partial_\mu Z^* Z) + \partial_\mu Z^* \partial_\mu Z$

$= (A_\mu + \frac{i}{2}(Z^* \partial_\mu Z - \partial_\mu Z^* Z))^2 + \partial_\mu Z^* \partial_\mu Z + \frac{1}{4}(Z^* \overleftrightarrow{\partial} Z)^2$

$Z^* \overleftrightarrow{\partial} Z$

$= \partial_\mu Z^* \partial_\mu Z + \frac{1}{4}(Z^* \overleftrightarrow{\partial} Z)^2$ $Z = \begin{pmatrix} \sqrt{1-|z|^2} \\ z_1 \\ z_2 \\ \vdots \end{pmatrix}$



$= \square F_{\mu\nu} F_{\mu\nu}$

ρ_μ ベクター場

↑ゲージ場 J.J. Sakurai 60

61 Nambu (π)

Chiral Theory

(3) Chiral Theory　6自由度　　QCD

$\pi^{\pm,0} \longrightarrow G/H = SU(2)_L \times SU(2)_R / SU(2)_V$

Linear 表現 $U = (2\times2)$ a matrix

{ ρ_μ → ゲージ理論
{ string 理論

8コの自由度

$g_{L,R} \in SU(2)_{L,R}$　G変換 $U \longrightarrow g_L U g_R^\dagger$　$\mathcal{L} = \frac{f_\pi^2}{4} Tr(\partial_\mu U \partial_\mu U^\dagger)$

$e^{i\vec{\theta}_{L,R}\vec{\sigma}}$　$\det U = \det U^\dagger = 1$　↓

$UU^\dagger = U^\dagger U = 1$　$U = e^{i\frac{\vec{\pi}(x)\sigma_i}{f_\pi}}$　$\frac{f_\pi^2}{4} Tr(g_L \partial_\mu U g_R^\dagger g_R \partial_\mu U^\dagger g_L^\dagger)$

$\pi_i \sigma_i = \begin{pmatrix} \pi_0 & \sqrt{2}\pi^+ \\ \sqrt{2}\pi^- & -\pi_0 \end{pmatrix}$　$f_\pi = 93\,MeV$　$g_L \partial_\mu U \partial_\mu U^\dagger g_L^\dagger$

(u.d.s)　$SU(3)_L \times SU(3)_R / SU(3)_V$　$= \mathcal{L}$

$U = \exp\left(i\frac{\hat{\pi}}{f_\pi}\right)$　$m_u \bar{u}u + m_d \bar{d}d + m_s \bar{s}s$

$\hat{\pi} = \lambda_a \pi^a \quad (a = 1\sim8)$　explicit Breaking term

$$= \begin{pmatrix} \frac{\eta}{\sqrt{3}} + \frac{\pi^0}{\sqrt{2}} & \pi^+ & K^+ \\ \pi^- & \frac{\eta}{\sqrt{3}} - \frac{\pi^0}{\sqrt{2}} & \bar{K}^0 \\ K^- & K^0 & -\frac{2}{\sqrt{6}}\eta \end{pmatrix}$$

$$\mathcal{L}_{CK} = (\bar{u}_R \ \bar{d}_R \ \bar{s}_R) \begin{pmatrix} m_u & 0 & \\ & m_d & \\ 0 & & m_s \end{pmatrix} \begin{pmatrix} u_L \\ d_L \\ s_L \end{pmatrix} = Q_L$$

$g_R^\dagger$　M　g_L

$Q_R \to g_R Q_R$

$\mathcal{L} = \frac{f_\pi^2}{4} Tr(\partial_\mu U^\dagger \partial_\mu U) + K\,Tr(MU + h.c)$　$Q_L \to g_L Q_L$　$M \to g_R M g_L^\dagger$

$m_{\pi^\pm}^2 = m_{\pi^0}^2 = K(x+y)$　$x = \frac{m_u}{m_s}$　$f_\pi = 0.248\ f_0 V^2$

$m_{K^+}^2 = K(x+1)$　$y = \frac{m_d}{m_s}$　$f_0 = 0.255\ GeV^2$

$m_{K^0}^2 = K(y+1)$

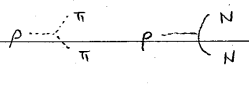
$m_\eta^2 = \frac{K}{3}(x+y+4)$

Gell-Mann, Okubo mass formura

$m_K^2 = \frac{3m_\eta^2 + m_\pi^2}{4}$

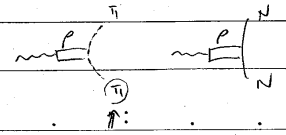
ρ meson as gauge field

1 ρ- coupling universality　$g_{\rho\pi\pi} = \square\, g_{\rho N\bar{N}} = G.$



2. $m_\rho^2 = 2 g_{\rho\pi\pi}^2 f_\pi$

3. ρ- dominance of the ploton coupling



Isospin = 1

Non-abelian gauge theory

$(\partial_\mu - ig T_a A^a_\mu)\psi$ ← $SU(2)$ ← Isospin $\frac{1}{2}i\tau_a$

$U(1)$ $(\partial_\mu - ieq A_\mu)\psi$

q = 定数

	q
ν	0
e	−1
p	+1
n	0

$SU(2)_L \times SU(2)_R / SU(2)_V$

$\begin{pmatrix} u \\ d \end{pmatrix} \to V \begin{pmatrix} u \\ d \end{pmatrix}$

Isospin → $A^\pm, A^0$ ⇑ $\rho^\pm$ ↑ ρ^0

Anderson (63)
Higgs Kibble (64)
ρ に mass を与える mechanism

(5) Higgs mechanism

$U(1)_{local}$: $\phi(x) \longrightarrow e^{i\theta(x)}\phi(x)$

$\mathcal{L} = D_\mu\phi^* D_\mu\phi - V(\phi) + \frac{1}{4e^2}F_{\mu\nu}F_{\mu\nu}$ $\quad D_\mu = \partial_\mu - iA_\mu$

$V(\phi) = -\mu^2|\varphi|^2 + \frac{\lambda}{2}|\phi|^4$ $\quad (\partial_\mu + iA_\mu)\phi^*(\partial_\mu - iA_\mu)\phi$

$\langle\phi\rangle = v = \sqrt{\frac{\mu^2}{\lambda}}$ $\quad A_\mu^2\langle\phi^*\phi\rangle = v^2 A_\mu^2$

$\phi = (v+\eta)e^{i\varphi}$

$\mathcal{L} = (v^2 + 2v\eta + \eta^2)(A_\mu - \partial_\mu\varphi)^2 + \partial_\mu\eta\,\partial_\mu\eta - V(\eta, v)$

"A_μ" $\equiv A_\mu - \partial_\mu\varphi$

$U(1)_{local}$ "A_μ" $\Rightarrow$ "A'_μ" $= (A_\mu - \partial_\mu\theta) - \partial_\mu\varphi + \partial_\mu\theta$ = "A_μ" は不変 "A_μ" は massive ベクトル場

$\langle\bar{q}q\rangle \neq 0$ ← 壊れていない

$SU(2)_L \times SU(2)_R \to SU(2)_V \to$ nothing

↑ ρ_μ ← mass を与えられる

massless vector 場 (+1, −1) スカラー 0×2

massive vector 場 (+1, 0, −1)

$\partial_\mu\varphi$ (η) $n=3$

physical Higgs 粒子

$SU(2)_{local} \to SU(2)_{global}$

$\varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix}$ $\varphi' = \begin{pmatrix} \varphi_1' \\ \varphi_2' \end{pmatrix}$

$\langle\varphi\rangle = \begin{pmatrix} 0 \\ v \end{pmatrix}$ $\langle\varphi'\rangle = \begin{pmatrix} v \\ 0 \end{pmatrix}$ $A_\mu^\pm, A_\mu^0$

$m^2_{A^\pm, A^0} = m^2$

Vector like theory

$q_{L\alpha}$ ↑ color $SU(3)$

$G/H = SU(2)_L \times SU(2)_R / SU(2)_V \simeq SU(2)_L \times SU(2)_R \times SU(2)_{local}$

$SU(2)_V$ ↑ ρ_μ ↑ ρ_μ

83 JJ. Sakurai

予定

1. ρ-meson ゲージ場
2. Weinberg-Salam Theory　　レポート
3. Baryon Number の破れ in the W-S Theory
4. 宇宙のバリオン数生成
5. 大統一理論

1960 J.J. Sakurai　$SU(2)_L \times SU(2)_R / SU(2)_V \simeq SU(2)_L \times SU(2)_R \times H_{local}$

$SU(2)_V$: iso spin symmetry

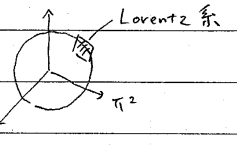
H_{local} = $SU(2)_H$ ゲージ場

Kugo. Uehara T.Y + Bando, Yamayama

$\mathcal{L}_{eff} = \frac{f_\pi^2}{4} Tr(\partial_\mu U \partial_\mu U^\dagger)$　ゲージ場

$U = \exp\left(\frac{2i\hat{\pi}}{f_\pi}\right)$　$\hat{\pi} = \frac{\sigma_i}{2}\pi^i$

$U^\dagger U = 1$, $\det U = 1$



Lorentz 座標

$g_{\mu\nu}(x) = e_\mu^i(x)\, e_\nu^i(x)$

一般座標

i については local な Lorentz 変換のもとで g 不変

$G_{ij}(\pi)$

$\mathcal{L} = \partial_\mu \pi^i G_{ij}(\pi) \partial_\mu \pi^j$

$SU(2)_L \times SU(2)_R$

$U(x)_{ij} \to g_L U(x) g_R^\dagger$　　$g_{L,R} \in SU(2)_{L,R}$

$U(x) \equiv (\xi_L^\dagger)_i^a (\xi_R)_j^a$　　$a = 1,2$　$i,j = 1,2$

local Hidden symmetry $(SU(2)_H)$　　$U = \xi^\dagger \xi$

$\xi_L(x) \Rightarrow h(x) \xi_L g_L^\dagger$　　$SU(2)_H$　$U \to \xi_L^\dagger h^\dagger(x) h(x) \xi_R = U(x)$

$\xi_R(x) \Rightarrow h(x) \xi_R g_R^\dagger$

$SU(2)_{local}$　　$h(x) = \exp(i\theta^i(x)\sigma^i)$

$\mathcal{L}(\xi_{L,R}\ D\ \xi_{L,R})$　　D: $\partial_\mu - ieA_\mu^i \sigma^i$

$\xi_L \to h(x) \xi_L$

$\xi_R \to h(x) \xi_R$

$D_\mu \xi_L \to h(x) D_\mu \xi_L$

$D_\mu \xi_R \to h(x) D_\mu \xi_R$

$SU(2)_L \times SU(2)_R \times SU(2)_H$

$\mathcal{L} = \mathcal{L}_A + \mathcal{L}_V$

$\mathcal{L}_A = -\frac{f_\pi^2}{4} Tr(D_\mu \xi_L \xi_L^\dagger - D_\mu \xi_R \xi_R^\dagger)^2$

$\mathcal{L}_V = -\frac{f_\pi^2}{4} Tr(D_\mu \xi_L \xi_L^\dagger + D_\mu \xi_R \xi_R^\dagger)^2$

$\xi_L(x) = e^{i\hat{\phi}(x)} \exp(-i\frac{\hat{\pi}}{f_\pi})$

$\xi_R(x) = e^{i\hat{\phi}(x)} \exp(-i\frac{\hat{\pi}}{f_\pi})$

$SU(2)_H$

Gauge fixing $\{\hat{\phi}=0\}$ $\xi_L^+ = \xi_R \equiv \xi = e^{i\frac{\hat{\pi}}{f_\pi}}$

$\xi^+\xi = 1$

$\mathcal{L}_A = \frac{f_\pi^2}{4} Tr\left\{\partial_\mu \xi^+ \xi - \partial_\mu \xi \xi^+\right\}^2 = \frac{f_\pi^2}{4} Tr\left\{\partial_\mu U \partial_\mu U^+\right\}$ $U = \xi^2$

$\mathcal{L}_V = f_\pi^2 Tr\left(\rho_\mu - \frac{1}{2i}(\partial_\mu \xi \xi^+ + \partial_\mu \xi^+ \xi)\right)^2$ (π)

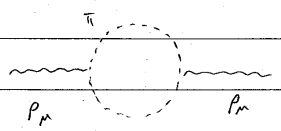
$\mathcal{L}_{kin} = \frac{1}{4g^2} F^{(\rho)}_{\mu\nu} F^{(\rho)}_{\mu\nu}$ $\rho \to m_\rho$ $g_{\rho\pi\pi}$ $m_\rho^2 \sim f_\pi^2 g_\rho^2$ ← $m_W = gf$

$\rho_\mu - \gamma$ の mixing

↓

mass less の γ

3 massive $\rho^\pm$ ρ^0



$\to$ $\log\Lambda\, F_{\mu\nu} F_{\mu\nu}$ ($\frac{1}{4g^2}$)

cut off

↓

$\Lambda = 1\,GeV$

$g_\rho^2 =$

$U = \xi_i^a \xi_{ai}$ (ξ が 2 ヶ所に入れる) $\rho_\mu \to$ 物理的になる $\Lambda = (\sim 100\,\pi)\left(\left[\square\right]\right)^{100}$

$U = e^{i\frac{2\hat{\pi}}{f_\pi}}$

IV Unification of Electromagnetic and Weak Interactions. Weinberg - Salam (1967)

Glashow (1961)

Weak interactions

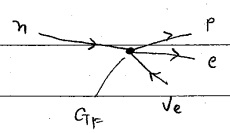
β-decay : $n \to p + e + \bar{\nu}_e$

$\mu \to \nu_\mu + e + \bar{\nu}_e$

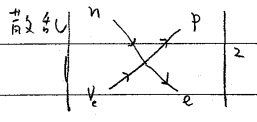
Fermi 理論 $H_{eff} = \frac{G_F}{\sqrt{2}} J_i J_i$ [3] $G_F \sim 10^{-5}\,GeV^{-2}$ $E = \int H_{eff}\, d^3x$

$J = \bar{p}\gamma_\mu(1-\gamma_5)n$

$j = \bar{e}\gamma_\mu(1-\gamma_5)\nu_e$



前ページ



$S = (E_{\nu_e} + E_n)^2 \sim 10^6\,GeV^2$

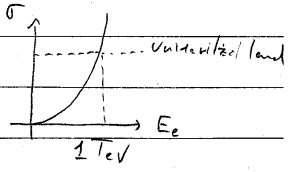
$E \sim \sqrt{S} \sim 10^3\,GeV$ で Unitarize が効き

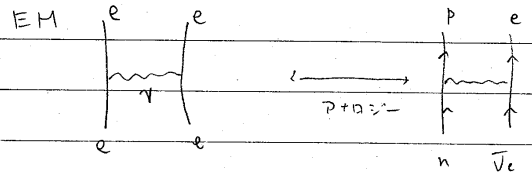
$E_{phys} < 1.6\,GeV$

$E < 1\,TeV$ で new physics が出現すべき

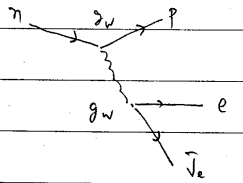
$\sigma(\nu_e n \to pe) = G_F^2 E_{\nu_e}^2$

面積 $= \frac{1}{E^2}$



EM 

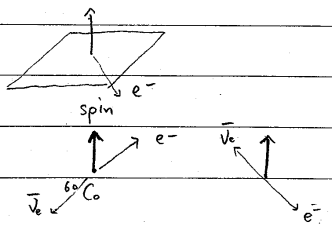
$m_W < 1\ TeV$



$$H_{eff} = \frac{g_W^2}{q^2 - m_H^2} \approx \frac{G_F}{\sqrt{2}} \quad \begin{cases} g_W \sim O(1) \\ m_W \simeq 1200\text{-}3000\ GeV \end{cases}$$

ニュートン 法則 | 地上の力学　　マックスウェル 電気力、磁気力
　　　　　　　| 天体の運動

$r \times v \to -r \times -v = r \times v$



(1) Electroweak Gauge Theory

(i) weak interaction

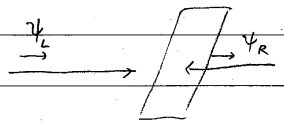
- Parity の破れ　　$P : (\vec{x}, t) \to (-\vec{x}, t)$　　$^{60}Co \to nuc + e + \bar{\nu}$

Lee-Yang　　$K \to 2\pi$　　$0^- \neq 0^+$ θ
　　0^-　　3π　　θ
　　0^+

- $m_W \neq 0$　自発的対称性の破れ

(ii) EM interactions

- Parity conserving　　$\psi_L \leftrightarrow \psi_R$



- $m_\gamma = 0$

$U(1)_{EM}$ がある

G (gauge 群)　　G の理論　ψ_L と ψ_R を区別する
↓　ψ_L と ψ_R は区別される　　ψ_L の G に対する変換性 ≠ ψ_R の (　　)

$U(1)_{em}$ のみ残る　　Lepton に注目
$\psi_R \leftrightarrow \psi_L$ 同等　　$\psi_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$　　$\psi_R = e_R$

最大な群　$SU(2)_L \times U(1)_L \times U(1)_R$

$SU(2)_L : \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \to e^{i\theta^i \theta^i} \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$

$U(1)_L : \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \to e^{i\alpha} \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$

$U(1)_R : e_R \to e^{i\beta} e_R$

$U(1)_{L+R}$ ← lepton 数に対応している　$Q_{Lepton} = \int d^3x : \bar{\psi}_L \gamma_\mu \psi_L + \bar{\psi}_R \gamma_\mu \psi_R :$

$[\psi_L, Q_{Lepton}] = \psi_L \quad [\psi_R, Q_{lepton}] = \psi_R$ ← global symmetry

弱い　③ ①　$SU(2)_L \times U(1)_Y$

local な gauge symmetry → ③+① のゲージ粒子 $W^\pm, Z^0, \gamma$

$U(1)_Y$ の charge Q_Y

$[Q_Y, \psi_L] = a\psi_L$ ⇐ $Q_Y|\psi_L\rangle = a|\psi_L\rangle$

$[Q_Y, e_R] = b e_R$ ⇐ $Q_Y|e_R\rangle = b|e_R\rangle$

$SU(2)_L \times U(1) \to U(1)_{em}$ （$U(1)$ 側）　$Q_{em} = I_3 + \frac{1}{2} Q_Y$

$I_3 \left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle = \begin{pmatrix}\frac{1}{2} & 0\\ 0 & -\frac{1}{2}\end{pmatrix}\left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle$

$Q_Y \left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle = \begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}\left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle$

$\begin{pmatrix}\nu_e\\ e\end{pmatrix}_L : (\frac{2}{2}, -1)$

$e_R : (1, -2)$

$Q_Y|e_R\rangle = b|e_R\rangle$

$Q_{em}|e_R\rangle = (-1)|e_R\rangle$

$b = -2$

$Q_{em}\left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle = \begin{pmatrix}0 & 0\\ 0 & -1\end{pmatrix}\left|\begin{pmatrix}\nu_e\\ e\end{pmatrix}\right\rangle$

$a = -1$

Higgs field　$\varphi = \begin{pmatrix}\varphi^+\\ \varphi^0\end{pmatrix}$　※ Higgs field を doublet とする ($\varphi = \begin{pmatrix}a\\ b\end{pmatrix}$)

$(\frac{2}{2}, +1)$

先ず $U(1)$ が残る それを $U(1)_{em}$ と identify する

$V = -\mu^2(\varphi^+\varphi) + \lambda(\varphi^+\varphi)^2$

$\langle\varphi\rangle = \begin{pmatrix}a\\ b\end{pmatrix} \to \begin{pmatrix}0\\ v\end{pmatrix}$

$SU(2)$ 回転

$\langle\varphi\rangle = \begin{pmatrix}0\\ v\end{pmatrix}/\sqrt{2}$

$v = \sqrt{\frac{\mu^2}{\lambda}}$

$\varphi = \begin{pmatrix}0\\ \eta\end{pmatrix} e^{i\xi'^{(x)}\sigma^i} \longrightarrow \begin{pmatrix}0\\ \eta\end{pmatrix} e^{i\xi^i\sigma^i + \theta'^{(x)}\sigma^i} = \begin{pmatrix}0\\ \eta\end{pmatrix}$

$SU(2)$ gauge 変換

gauge 粒子のうち4つのゲージ粒子のうち

3つが massive

1つが massless

$\mathcal{L} = \mathcal{L}_{Gauge\ field} + \mathcal{L}_{lepton} + \mathcal{L}_{Higgs}$

$\mathcal{L}_{Gauge} = -\frac{1}{4} F^i_{\mu\nu} F^i_{\mu\nu} - \frac{1}{4} B_{\mu\nu} B_{\mu\nu}$　($SU(2)_L$, $U(1)_Y$, $i = 1-3$)

$F_{\mu\nu} = \partial_\nu A^i_\mu - \partial_\mu A^i_\nu + g\epsilon^{ijk} A^j_k A^k_\nu$

$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$

$\mathcal{L}_{lepton}$: e_R — g' — B_μ — e_R

$\mathcal{L}_{Higgs} = (D_\mu\varphi)^+ D_\mu\varphi - V(\varphi^+\varphi)$

$= \varphi^+\left(\partial_\mu + \frac{ig'}{2}B_\mu + i\frac{g}{2}\sigma^i A^i_\mu\right)\left(\partial_\mu - i\frac{g'}{2}B_\mu - i\frac{g}{2}\sigma^i A^i_\mu\right)\varphi$

$\begin{pmatrix}\nu\\ e\end{pmatrix}_L = l_L$ — $g\sigma^i$ — W^i_μ — l_L

gauge field の mass term　$\mathcal{L}_{mass} = \frac{1}{8}(v)^2\left[(g'B_\mu - gA^3_\mu)^2 + g^2((A'^i_\mu)^2 + (A^2_\mu)^2)\right]$

A_μ が massless ← photon　$= \frac{v^2}{4}g^2 W^+W^- + \frac{1}{2}v^2\frac{g^2+g'^2}{4}(Z^0_\mu)^2$

$W^\pm_\mu = \frac{A^1_\mu \pm A^2_\mu}{\sqrt{2}}$

$Z^0_\mu = \frac{gA^3_\mu - g'B_\mu}{\sqrt{g^2+g'^2}}$　$A_\mu = \frac{gB_\mu + g'A^3_\mu}{\sqrt{\quad}}$　$\tan\theta_W = \frac{g'}{g}$

$\tan\theta_W = \frac{g'}{g}$ $\quad e = g\sin\theta_W$ $\quad m_W^2 = \frac{e^2}{\sin^2\theta_W}\cdot\frac{1}{4\sqrt{2}G_F}$

$e = \frac{gg'}{\sqrt{g^2+g'^2}}$ $\quad g = \frac{e}{\sin\theta_W}$

$m_Z = \frac{m_W}{\cos\theta_W}$ $\quad \sin^2\theta_W = 0.23 \rightarrow m_W = 76\,GeV \rightarrow$ 実験 $80.65\,GeV$

$m_Z = 89\,GeV$ 5% $91.1\,GeV$

$\mathcal{L}_{Yukawa} = \lambda_e \bar{e}_R l_L \langle\varphi\rangle + h.c$

$= \lambda_e \frac{v}{\sqrt{2}} \bar{e}_R e_L + h.c.$ (m_e)

quarkへの拡張 → $\begin{pmatrix} u \\ d \end{pmatrix}_L$ u_R, d_R

fermions

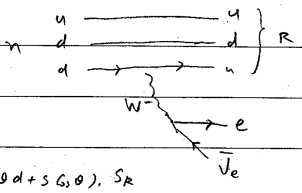
1st family	$\begin{pmatrix}\nu_e\\ e\end{pmatrix}_L$ e_R	$\begin{pmatrix}u\\ d\end{pmatrix}_L$ u_R e_R	15	e 0.5MeV, $m_u \sim 3$MeV, $m_d \sim 5$MeV
2nd 〃	$\begin{pmatrix}\nu_\mu\\ \mu\end{pmatrix}_L$ μ_R	$\begin{pmatrix}c\\ s\end{pmatrix}_L$ c_R s_R	15	$m_c \sim 1$GeV, $m_s \sim 300$MeV
3rd 〃	$\begin{pmatrix}\nu_\tau\\ \tau\end{pmatrix}_L$ τ_R	$\begin{pmatrix}t\\ b\end{pmatrix}_L$ t_R b_R	15	$m_t \sim 170$GeV, $m_b \sim 4.5$GeV

$SU(2)\times U(1)$ → $W^\pm$ Z^0 A_μ (中性カレント)

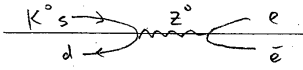
$\mathcal{L} = W^-_\mu J^+_\mu + Z_0 J^0_\mu + A_\mu J_{em}$

$\mathcal{L} = (\bar{d}) \gamma_\mu d_L Z^0_\mu + \bar{u}\gamma_\mu d\, W^+$

$\begin{pmatrix} u \\ d\cos\theta + s\sin\theta \end{pmatrix}$, u_R, d_R, $(-\sin\theta d + s\cos\theta)$, s_R

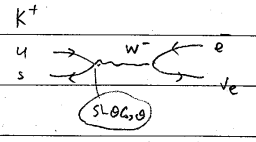


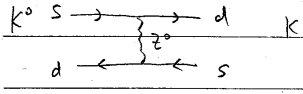
FCNC problem



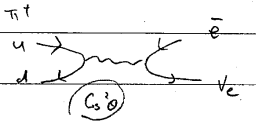
$(\cos\theta d_L + \sin\theta s_L)^2 Z^0_\mu$

$\sin\theta\cos\theta(\bar{d}\gamma_\mu s_L + \bar{s}_L \gamma_\mu d_L) Z^0_\mu$





K^0 mixing $\quad G_{eff} = \begin{pmatrix}10^{-9}\\ 10^{-15}\end{pmatrix} G_F$



Jeorgi - Glashow

SO(3) $\begin{pmatrix} u \\ \overline{\cos\theta d + \sin\theta s} \\ \times \end{pmatrix}$ $W^\pm$ $\quad (\bar{u}u - \bar{x}x) Z^0_\mu$ $\quad T_3 = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}$

Glashow Iliopoulos Maiani $\quad \Sigma_0(\bar{u}u - \bar{d}d + \bar{c}c - \bar{s}s)$

Maki Hama

cとそちこそ $\quad \begin{pmatrix}u\\ d\end{pmatrix}\begin{pmatrix}c\\ s\end{pmatrix} \rightarrow \begin{pmatrix}u\\ \cos\theta d + \sin\theta s\end{pmatrix}\begin{pmatrix}c\\ -\sin\theta d + \cos\theta s\end{pmatrix}$

V. Baryon-Number Violation in the Weinberg-Salam Model

$\mathcal{L} =$ gauge field kinetic term $\{F_{\mu\nu}F_{\mu\nu} + \cdots\}$ $+ G_e \bar{e}_R l_L \varphi + h.c.$ $\quad l_L = \begin{pmatrix}\nu_e\\ e\end{pmatrix}_L$ $\quad q_L = \begin{pmatrix}u\\ d\end{pmatrix}_L$

$+ \bar{l}_L \not{D}_l l + \bar{e}_R \not{D}_e e_R$ $+ G_d \bar{d}_R q_L \varphi + h.c.$

$+ \bar{q}_L \not{D}_q q_L + \bar{u}_R \not{D}_u u_R + \bar{d}_R \not{D}_d d_R$ $+ G_u \bar{u}_R q_L \varphi^* + h.c.$

$- V(\varphi^+\varphi)$

Lepton数 保存する

Baryon数 $P = \frac{1}{3}$

保存カレント $\partial_\mu J^i_\mu = 0$

保存数 $Q_i = \int J^i_0 d^3x$

$\dot{Q}_i = 0 \quad [Q_i, Q_j] = 0$

$j^L_\mu = (\bar{l}_L \gamma_\mu l_L + \bar{e}_R \gamma_\mu e_R)$

$j^B_\mu = \frac{1}{3}(\bar{q}_L \gamma_\mu q_L + \bar{u}_R \gamma_\mu u_R + \bar{d}_R \gamma_\mu d_R)$

$\partial_\mu j^L_\mu = \partial_\mu j^B_\mu = 0$ ← 古典論

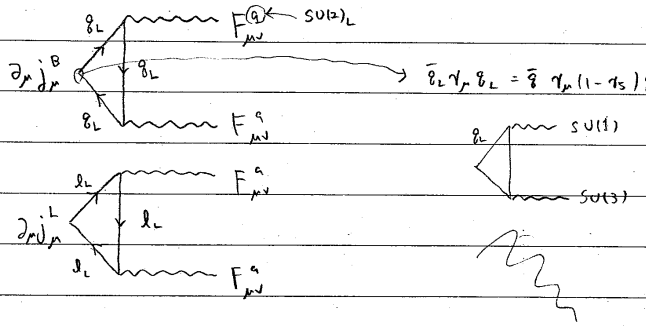
U(1) anomalies

$\partial_\mu j^B_\mu = \frac{N_f}{32\pi^2} g^2 F^a_{\mu\nu} \tilde{F}^a_{\mu\nu}$

$\partial_\mu j^L_\mu = \frac{N_f}{32\pi^2} g^2 F^a_{\mu\nu} \tilde{F}^a_{\mu\nu}$

量子論

$\tilde{F}^a_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma} F^a_{\rho\sigma}$



B−L は保存する

$\partial_\mu (j^B_\mu - j^L_\mu) = 0$

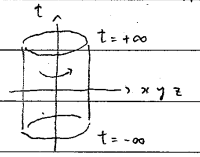
量子論でも

$\int d^4x\, \partial_\mu j^B_\mu = \frac{g^2}{32\pi^2} \int d^4x\, F^a_{\mu\nu} \tilde{F}^a_{\mu\nu} \quad (a = 1 \sim 3)$

$\parallel$

$\int d\sigma_\mu j^B_\mu$

Surface



$= \int_{t=+\infty} d^3x\, j^B_0 - \int_{t=-\infty} d^3x\, j^B_0 + \int_{R=\infty} d\vec{s}\, \vec{j}^B_s$

$= Q_B(t=+\infty) - Q_B(t=-\infty) \neq 0$

バリオン数の非保存

(1976) t'Hooft

$P = |e^{-S(\text{instantion})}|^2$

$\sim e^{-\frac{16\pi^2}{g^2}} \simeq 10^{-170}$

は Strongly に Suppress される

1985~6

(2) Vacua in the W-S Model

$SU(2)_L \times U(1)_Y$

$\mathcal{L} = -\frac{1}{4} F^a_{\mu\nu} F^a_{\mu\nu} + (D_\mu \varphi)^+ (D_\mu \varphi) - V(\varphi)$

$D_\mu \varphi = (\partial_\mu - \frac{ig}{2} \sigma^a W^a_\mu) \varphi$

$V(\varphi) = \lambda(\varphi^+ \varphi - \frac{1}{2} v^2) \Rightarrow \varphi = \begin{pmatrix} 0 \\ v \end{pmatrix} / \sqrt{2}$

$F^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + \cdots$

$\mathcal{L} \to H$

$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} \to \mathcal{H} = \pi \dot{\varphi} - \mathcal{L}$

$\dot{W}^a_0$ が $\mathcal{L}$ の中にない

系が W_0 に indep

$\to W^a_0 = 0$ の系に fix する

$\to \mathcal{L} = \frac{1}{2}(\dot{W}_i)^2 - \frac{1}{4} F^a_{ij} F^a_{ij} + |\dot{\varphi}|^2 - |D_\mu \varphi|^2 - V(\varphi)$

$i = 1 \sim 3$

$\pi^a_{W_i} = \frac{\partial \mathcal{L}}{\partial \dot{W}^a_i} = W^a_i$

$\pi_{\varphi^*} = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}^*} = \dot{\varphi}$

$\mathcal{H} = \pi_{W_i} \dot{W}_i + \pi_\varphi \dot{\varphi} + \pi_{\varphi^*} \dot{\varphi}^* - \mathcal{L} = (\dot{W}_i)^2 + 2|\dot{\varphi}|^2 - \mathcal{L}$

$= \frac{1}{2}(\dot{W}_i)^2 + |\dot{\varphi}|^2 + \frac{1}{4}(F^a_{\mu\nu})^2 + |D_\mu \varphi|^2 + V(\varphi)$

No No No No No

Vacua : grand states

W_i
φ

$\left\{\begin{array}{l} \dot{W}_i = 0 \\ \dot{\varphi} = 0 \end{array}\right\}$ 真空は static である

$\left\{\begin{array}{l} F^a_{ij} = 0 \\ D_\mu \varphi = 0 \\ V(\varphi) = 0 \end{array}\right\}$ ①

$F = \partial W + \partial W + [W, W]$

$W = 0 \rightarrow W = \partial_\mu U U^{-1}$

Vacuum configuration

$W_i(\vec{x})$, $\varphi(\vec{x})$ $\quad \vec{x} \in R^3$

R^3 の全空間で ① をみたすもの

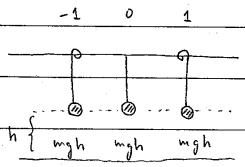
$\rightarrow W_i(\vec{x})$ は pure gauge である

$\left\{\begin{array}{l} \hat{W}_i(\vec{x}) \equiv \sigma^a W_i^a = -\frac{2i}{g} \partial_i g(\vec{x}) \cdot g^{-1}(\vec{x}) \\ \varphi(\vec{x}) = \frac{v}{\sqrt{2}} g(\vec{x}) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{array}\right.$ $\quad e^{i\theta_i \sigma^i}$ $\quad g(\vec{x}) \in SU(2)_L$

$\varphi^\dagger \varphi = \frac{v^2}{2} (0\ 1)\, g^\dagger g \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{v^2}{2}$

$V(\varphi) = 0$

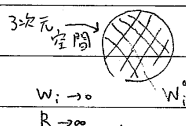
$D_\mu \varphi = \partial_i \varphi - \frac{1}{2} g W_i \varphi = \left\{ \partial_\mu g(x) + \frac{1}{2} g \frac{-2i}{g} \partial_i g(x) g^{-1}(x) g(x) \right\} \frac{v}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$



真空が無限に縮退している

$g(x)$ の自由度

$e^{i\theta_i(x)\sigma^i}$



無限遠点を同一視する $\quad R^3 \rightarrow S^3$

$g(\vec{x}) : S^3 \rightarrow SU(2)$ mapping

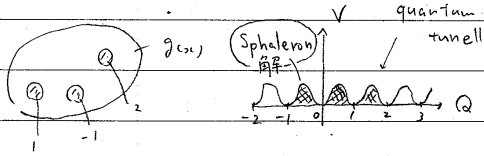
g をトポロジカルに分類する

$\Pi_3(SU(2)) = Z_n$

topological charge

$Q_T \equiv \frac{1}{24\pi^2} \int d^3x\, \epsilon^{ijk}\, Tr\left(\partial_i g g^{-1} \partial_j g g^{-1} \partial_k g g^{-1}\right)$

$= \text{integer}$



$Q_T = 0 \rightarrow Q_T = 1$

$Q_B = n \rightarrow Q_B = n+1$

$\partial_\mu j^B_\mu = \frac{g^2}{32\pi^2} F_{\mu\nu} \tilde{F}_{\mu\nu}$ $\qquad K_\mu = \frac{g^2}{32\pi^2} \epsilon_{\mu\nu\rho\sigma} \left\{ F^a_{\nu\rho} W^a_\sigma - \frac{1}{3} g \epsilon^{abc} W^a_\nu W^b_\rho W^c_\sigma \right\}$ $\qquad Q_c = \int d^3x\, K_0$

$\int d^4x\, \partial_\mu j^B_\mu = \frac{g^2}{32\pi^2} \int d^4x\, F_{\mu\nu} \tilde{F}_{\mu\nu} = Q_c(+\infty) - Q_c(-\infty)$

$Q_B(+\infty) - Q_B(-\infty)$ $\qquad \partial_\mu K_\mu$ $\qquad \Delta Q_B = \Delta Q_c$

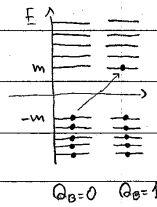
$\Delta(Q_B - Q_c) = 0$

$Q_c(W_\mu(\cdot)) = Q_T$ $\rightarrow$ $\boxed{\Delta(Q_B - Q_T) = 0}$

真空解 (g で書く)

Dirac 理論

$(\not{D}_W - m)\psi = 0$



$\gamma^\mu (\partial_\mu - i g W^a_\mu \sigma^i) \psi = 0$

$\partial_\mu g(x) g^{-1}$ $\quad Q_T = n$

$\partial_\mu g'(x) g'^{-1}(x)$ $\quad Q_T = n+1$

$P \sim e^{-\frac{M}{T}}$

✠ Aberian-Higgs Theory in (1+1) dimension

U(1)

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F_{\mu\nu} + (D_\mu \varphi)^* (D_\mu \varphi) - \frac{\lambda}{2} (\varphi^* \varphi - 1)^2$$

$$\begin{cases} F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu . \\ D_\mu \varphi = (\partial_\mu - i A_\mu) \varphi \end{cases}$$

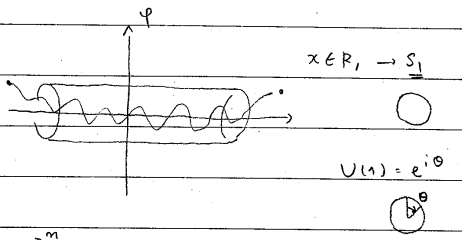
(i) Topological Vacua $A_0 = 0$ gauge に fix する

$$\mathcal{L} = \frac{1}{2} (\dot{A}_1)^2 + |\dot{\varphi}|^2 - (D_1 \varphi)^* (D_1 \varphi) - \frac{1}{2} \lambda (\varphi^* \varphi - 1)^2$$

$$\downarrow$$

$$H = \frac{1}{2} (\dot{A}_1)^2 + |\dot{\varphi}|^2 + |D_1 \varphi|^2 + \frac{1}{2} \lambda (\varphi^* \varphi - 1)^2$$

$\dot{A}_1 = 0$, $\dot{\varphi} = 0$ (static), $D_1 \varphi = 0$



↓ $A_1(x)$ $\varphi(x)$

Vacuum solutions

$\varphi = U_n(x)\, e^{i\theta_n}$ $\qquad \varphi^+ \varphi = U_n^+ U = 1$ $\qquad U_n(x) = \left[\frac{\alpha - ix}{\alpha + ix}\right]^n$ $(\alpha > 0)$

$A_1 = -i \partial_1 U_n(x) \cdot U_n(x)^{-1}$ $\qquad D_1 \varphi = 0$ $\qquad \varphi(x) \longrightarrow (-1)^n$ $(x = \pm\infty)$

The Chern Simons Charge Q_c (真空) $= Q_T$

~~$Q = \frac{1}{2\pi} \int dx\, A_1(x) = -i \partial_1 g g^{-1}$~~

$$\partial_1 U_n(x) = n \left(\frac{\alpha - ix}{\alpha + ix}\right)^{n-1} \frac{-2i\alpha}{(\alpha + ix)^2}$$

$$-i\, \partial_1 U_n(x) \cdot U_n^{-1}(x) = -2n\alpha \frac{1}{\alpha^2 + x^2}$$

$$Q_T = \frac{-2n\alpha}{2\pi} \int_{-\infty}^{\infty} \frac{1}{x^2 + \alpha^2} dx = -\frac{n}{\pi} \underbrace{\operatorname{Arctan} \frac{x}{\alpha}\Big|_{-\infty}^{\infty}}_{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)} = -n$$

Q → gauge 変換を受ける」

$\varphi \to e^{i\beta(x)} U_n(x)$

$A_1 \to i \partial_1 U_n(x) U_n^{-1}(x) + \partial_1 \beta(x)$

$Q \to \frac{1}{2\pi} \int A_1 dx = n + \frac{1}{2\pi} \int dx\, \partial_1 \beta(x) = n + \beta(+\infty) - \beta(-\infty)$

しかし $\Delta Q = Q(t = +\infty) - Q(t = -\infty) = n - n'$

はゲージ不変量

✠ Sphaleron 解

static, classical field equation の解 $\dot{\varphi} = 0$

$$\begin{cases} A_1 = \partial_1 \beta(x) \\ \varphi = e^{i\beta(x)} \varphi(x) \end{cases} \quad \varphi(x) \text{ real}$$

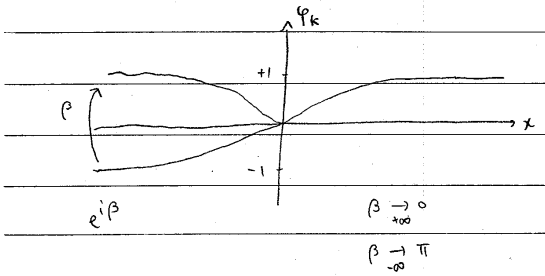
$(D_1)^2 \varphi = \lambda (\varphi^* \varphi - 1) \varphi$

$D_1 \varphi = (\partial_1 - i A_1) \varphi = (i \partial_1 \beta e^{i\beta} \varphi + e^{i\beta} \partial_1 \varphi - i \partial_1 \beta e^{i\beta} \varphi) = e^{i\beta} \partial_1 \varphi$

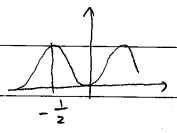
$(D_1)^2 \varphi = e^{i\beta(x)} \partial_1^2 \varphi(x) = \lambda e^{i\beta} (\varphi^2 - 1) \varphi$

$$\boxed{\partial_1^2 \varphi(x) = \lambda (\varphi^2 - 1) \varphi}$$ kink solution

$\varphi_K = \tanh\left(\frac{\sqrt{\lambda}}{2} x\right)$



$Q_{sph} = \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\, A_1$ $\quad Q_{Vacuum} = \text{integer}$ $\quad Q_{sph} = \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\, \partial_1 \beta(x) = \frac{1}{2\pi}[\beta(+\infty) - \beta(-\infty)] = -\frac{1}{2}$



(1+3)D Weinberg-Salam Model

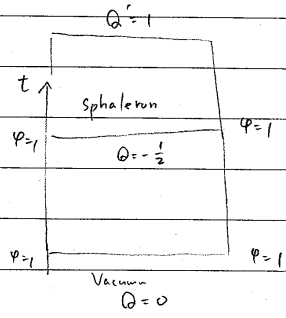
(SU(2) - Higgs Model)

$(x,y,z,t) \to (r,t)$

$W_j^a(\vec{x},t) = \frac{1}{g}\{A(r,t)\,\epsilon_{jam} r_m + B(r,t)[r^2\delta_{ja} - r_j r_a] + C(r,t)\, r_j r_a\}$

運動方程式

$W_0^a = \frac{1}{g} G(r,t)\, r_a / r$



~~$\varphi(x) = \frac{v}{\sqrt{2}}\{H(r,t) + K(r,t)\frac{i\vec{\tau}\cdot\vec{r}}{r}\}\begin{pmatrix}0\\1\end{pmatrix}$~~

$\mathcal{L}(W,\varphi) \to \mathcal{L}_{eff} = \frac{4\pi}{g}\left\{-\frac{1}{4}r^2 f_{\mu\nu}f_{\mu\nu} + |Dx|^2 + r^2|D_\mu\varphi|^2 - \frac{1}{2r^2}(|x|^2-1)^2 - \frac{\lambda}{g^2}r^2(|\varphi|^2 - \frac{1}{2}g^2v^2)^2\right.$

$\left. -\frac{1}{2}(|x|^2+1)|\varphi|^2 - \mathrm{Re}(ix^*\varphi^2)\right\}$

$L = \int dr\, \mathcal{L}_{eff}$

$f_A = 1 + r^2 A$ $\quad \begin{pmatrix}a_0\\a_i\end{pmatrix} = \begin{pmatrix}G\\f_c\end{pmatrix}$ $\quad$ local U(1) transformation

$f_B = r^3 B$ $\quad x \to e^{i\theta(x,t)}x$ $\quad D_\mu x = (\partial_\mu - ia_\mu)x$

$f_c = r^2 c$ $\quad \varphi \to e^{i\frac{\theta(x,t)}{2}}\varphi$ $\quad D_\mu\varphi = (\partial_\mu - \frac{i}{2}a_\mu)\varphi$

$\varphi = \sqrt{2}m_w(H+ik)$ $\quad a_\mu \to a_\mu + \partial_\mu\theta(x,t)$ $\quad \Rightarrow (1+1) = \Sigma\pi$

$-ix = f_A + if_B$ $\quad$ Abelian Higgs Model

$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu \quad (\mu = 0,1)$

$H_{eff} = \frac{4\pi}{g^2}\left\{0\,|\dot{a}|^2 + 0\,|\dot{x}|^2 + 0\,|\dot{\varphi}|^2 + |D_1 x|^2 + r^2|D_1\varphi|^2 + \frac{1}{2r^2}(|x|^2-1)^2 + \frac{\lambda}{g^2}r^2(|\varphi|^2 - 2m_w^2)^2 + \frac{1}{2}(|x|^2+1)|\varphi|^2 + \mathrm{Re}(ix^*\varphi^2)\right\}$

真空

	Solution
$D_1 x = D_1\varphi = 0$	
$\lvert x\rvert^2 = 1$	$a_1(r) = \partial_1\theta(r)$
$\lvert\varphi\rvert^2 = 2m_w^2$	$x(r) = -i e^{i\theta(r)}$
$(\lvert x\rvert^2+1)\lvert\varphi\rvert^2 = -2\mathrm{Re}(x^*\varphi^2)$	$\varphi/\sqrt{2}m_w = \exp(\frac{i}{2}\theta(r))$

The chern Sinons Charge

$Q = \frac{g^2}{32\pi^2}\int d^3x\, \epsilon^{ijk}\left\{F_{ij}^a W_k^a - \frac{2}{3}g\epsilon^{abc}W_i^a W_j^b W_k^c\right\}$

$= \frac{1}{2\pi}\int_0^\infty dr\left\{a_1 + \frac{1}{2}i[(D_1x)^2 x - x^*(D_1x)]\right\} = \frac{1}{2\pi}\{\theta(r=\infty) - \theta(r=0)\}$

$\varphi(r) \xrightarrow[r\to\infty]{} 1$

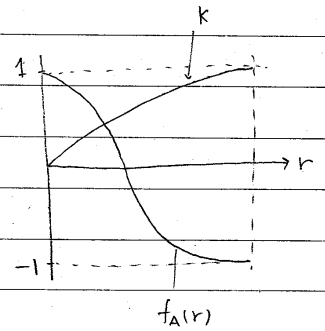
$\theta(r) \xrightarrow[r\to\infty]{} 0$

$W_j = \square + \frac{1-\cos\theta(r)}{r^2}\epsilon_{jam} r_m$ $\boxed{\theta(0) = -2n\pi}$ $Q = \frac{1}{2\pi}(0 - 1 - 2\pi n)) = n$

$W_j \xrightarrow[r\to 0]{} \text{regular}$

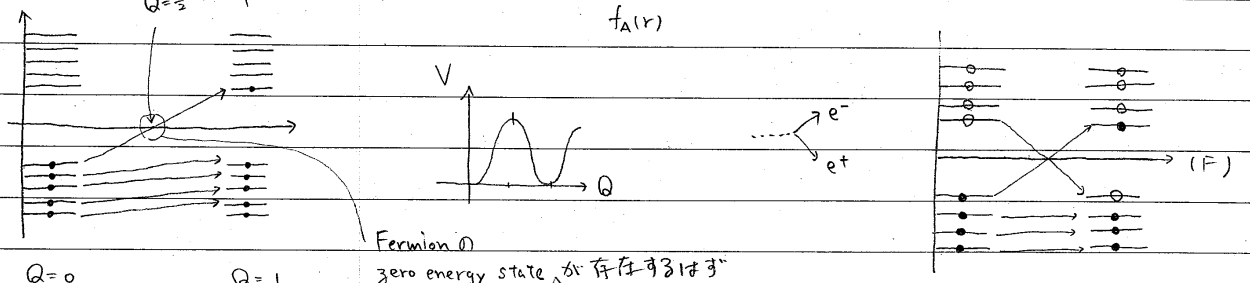
Sphaleron 解 (kink 解)

$\begin{cases} f_B = H = 0 \\ f_A(r) = 1 - f(\xi) \\ K(r) = f_h(\xi) \end{cases}$



$\xi = 2m_W r$

$\begin{cases} \varphi = 0 \quad (r=0) \\ \quad = \sqrt{2}m_W \quad (r=\infty) \\ \chi = +i \quad (r=0) \\ \quad = -i \quad (r=\infty) \end{cases}$



Fermion

$D_\mu \gamma_\mu \psi_L = 0$ $\psi(x) = \left[f(r,t) + ig(r,t)\frac{\vec{r}\cdot\vec{\sigma}}{r}\right]\frac{\xi}{r}$ $\xi = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $\eta = \begin{pmatrix} f \\ g \end{pmatrix}$

$(\partial_\mu - ig(W_\mu^i)\tau^i)$

↑ Sphaleron 解を入れる

$\mathcal{L} = \bar{\psi}\not{D}\psi \to \mathcal{L}_{eff}(f, g, W)$

$\left(i\sigma_1\partial_0 - \sigma_3\frac{\partial}{\partial r} + \frac{f_A(r)}{r}\right)\eta(r,t) = 0$ at the sphaleron

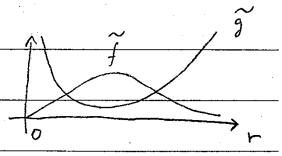
$\eta(r,t) = \begin{pmatrix} \tilde{f}(r) \\ \tilde{g}(r) \end{pmatrix} e^{-iEt}$ $(f_B = 0)$

$\begin{cases} \frac{d\tilde{f}}{dr} - \frac{f_A(r)}{r}\tilde{f} = E\tilde{g} \\ \frac{d\tilde{g}}{dr} + \frac{f_A(r)}{r}\tilde{g} = E\tilde{f} \end{cases}$ $(E=0)$

$\tilde{f}(r) = c\exp\left[\int^r f_A(r')\frac{dr'}{r'}\right]$

$\xrightarrow[r\to\infty]{} \sim c\exp\left[\int^r \frac{dr'}{r'}\right] \sim \frac{c'}{r}$

$\xrightarrow[r\to 0]{} c\exp\left[\int^r \frac{dr'}{r'}\right] \sim c''r$



$\tilde{g} \xrightarrow[r\to\infty]{} r$

$\xrightarrow[r\to 0]{} \frac{1}{r}$

VI 宇宙の Baryon 数

(1) Baryon 数の問題

我々の宇宙 $N_B \gg N_{\bar{B}}$

- 宇宙線の成分

  反物質 / 物質 $< 10^{-4}$

- γ の excess

  $N + \bar{N} \longrightarrow \pi^0 + \cdots$

  $\pi^0 \to 2\gamma$ $\langle E_\gamma \rangle \sim 200$ MeV

  No evidence

  $\frac{\text{反物質}}{\text{物質}} \lesssim 10^{-5} \sim 10^{-10}$ (銀河内)

1967年 Sakharov

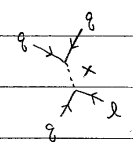
$N_B = 0$ 出発 $\longrightarrow$ $N_B \neq 0$ 宇宙

条件) 1. Baryon 数非保存

2. CP の破れ ← K の CP violation の発見の直後

3. 非平衡過程

1. 高温の初期宇宙を考える



$m_X \gg m_p$

$\frac{1}{m_X^2}$

ただし高温では

$X \to qq$ $(2/3)$

$\to \bar{q}\, l$ $(-1/3)$

GUT

Proton は Decay する $p \left(\begin{matrix} q \to \\ q \to \\ q \end{matrix} \right) \pi$

l, e^+

$p \to \pi^0 + e^+$

$\Rightarrow$ ν が mass をもつ

太陽ニュートリノ問題

$X \longrightarrow \underset{3\ 3}{qq}$ $(2/3)$

$\longrightarrow \underset{3^*}{\bar{q}\, l}$ $(-1/3)$

$\square_3 \times \square_3 = \underset{3^*}{\square\!\square} + \square\!\square$

3×3

$\parallel$

3^*

$O(3)$

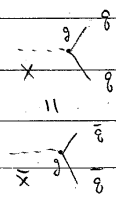
$(L \times L)_{3\ 3}$, $L = 1$

$\mathcal{L} = \lambda_1 \underbrace{\bar{q}\bar{q}X}_{3^*\,3^*\,3^*} + \lambda_2 \underbrace{q\bar{l}X}_{3\,1\,3^*}$

$\mathbb{1}$ $\mathbb{1}$

$Br(X \to qq) > Br(X \to \bar{q}l)$

物質 > 反物質



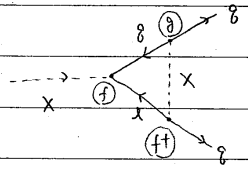
Born 近似では等しい

$\#qq = \#\bar{q}\bar{q}$

$\Gamma_{X\to qq} - \Gamma_{\bar{X}\to\bar{q}\bar{q}} \neq 0$

CPT $\begin{cases} m_X = m_{\bar{X}} \\ \Gamma_X = \Gamma_{\bar{X}} \end{cases}$ $\Gamma_{X\to qq} = |g|^2 = \Gamma_{\bar{X}\to\bar{q}\bar{q}}$

$\Gamma_X = \Gamma_{X\to qq} + \Gamma_{X\to\bar{q}\bar{q}}$



$\Gamma_{X\to qq} - \Gamma_{\bar{X}\to\bar{q}\bar{q}}$

$\propto |\sum_i G_i f_i(s)|^2 - |\sum_i G_i^* f_i(s)|^2$ $\quad g\,f\,f^+$ / G

$= \sum_{ij} G_i G_j^* f_i(s) f_j^*(s) - \sum_{i,j} G_i^* G_j f_i(s) f_j^*(s)$

$= 2i \sum_{i,j} \mathrm{Im}(G_i G_j^*) f_i(s) f_i^*(s) = i \sum_{i,j} \mathrm{Im}(G_i G_j^*) \{ f_i(s) f_j^*(s) - f_j(s) f_i^*(s) \}$

$= -2\,\mathrm{Im}(G_i G_j^*)\,\mathrm{Im}\, f_i(s) f_j^*(s) \neq 0$

$\mathrm{Im}\, f_i(s) f_j^*(s) \neq 0$

◀ CP violation

$\begin{pmatrix} +1 \\ 0 \\ -1 \end{pmatrix}$ $O(3)$

also $O(2)$

(2) 大統一理論 1972, 3年 Georgi - Glashow

Standard model

$SU(3) \times SU(2) \times U(1)$

$\begin{pmatrix} \nu \\ e \end{pmatrix}_L$, e_R $\begin{pmatrix} u \\ d \end{pmatrix}_L$ d_R u_R $= 15$

Q_{em} : $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$ -1 $\begin{pmatrix} 2/3 \\ -1/3 \end{pmatrix}$ $-1/3$ $2/3$ (4×3)

Charge の量子化

$U(1)$ の charge は実数

→ 0, −1, 2

$G \supset U(1)$

$SU(5)_{GUT}$

$A_{adj} = \left(\begin{array}{c|c} 3\times3 & \\ \hline & 2\times2 \end{array}\right)$ (SU(3) × SU(2)) — $U(1)_Y$

$15 = 5^* + 10$

↑ Observation

$Q_Y = \begin{pmatrix} 2 & 2 & 2 & & \\ & & & -3 & -3 \end{pmatrix}$

$Q_Y 5^* = \begin{pmatrix} +\frac{2}{3} & +\frac{2}{3} & +\frac{2}{3} & & \\ & & & -1 & -1 \end{pmatrix} \begin{pmatrix} \bar{q} \\ \bar{q} \\ \bar{q} \\ l \\ l \end{pmatrix}$ anti quark ↓ ; ↑ doublet

$Q_{em} = \begin{pmatrix} 1/3 & 1/3 & 1/3 & & \\ & & & 0 & -1 \end{pmatrix} \begin{pmatrix} \bar{d} \\ \bar{d} \\ \bar{d} \\ \nu \\ e \end{pmatrix}$

$Q_{em} = I_3 + \frac{1}{2} Y$

$\begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$

$10 = \left(\begin{pmatrix} u \\ d \end{pmatrix} \bar{u} \cdot \bar{e} \right)$

$\bar{u}$ ↓ u_R ; $\bar{e}$ ↓ e_R

gauge 粒子 = Adj = 3 × | gluons | X(3) | B_μ
| $\bar{X}(3^*)$ | $W^\pm W^0$ |

$X \to \bar{q} + \bar{q}$ $(\Delta B = -2/3)$ $\Delta(B-L) = -2/3$
$\to q + \bar{l}$ $(\Delta B = +1/2)$ $\Delta(B-L) = -2/3$

? Proton Decay $q \to \pi$, $q \to \pi$, $q \to \pi + l$

$P \to \pi^0 + e^+$ //

CP 発見
↓
Saharov B-genesis
↓
GUT support
Proton decay ?

@ α^{-1} : U(1), SU(2), SU(3)

B-L は保存する

$T \sim 10^{16}$ GeV
10^{14} GeV
Baryon 生成

$\frac{d}{dt}(B-L) = 0$

$B(t) = \frac{B(t)+L(t)}{2} + \frac{B(t)-L(t)}{2}$

$t \to \infty$ ↓

$B(\infty) = \left(\frac{B-L}{2}\right)_{initial}$

B = 0, L = 0
B-L = 0, B+L = 0
Corriction

◀

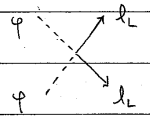
VI ⇒ (i) (B-L) の破れが必要 ⇒ ν が mass をもつ
(i$_0$) B の破れが必要 ⇒ 陽子が崩壊する

B-L の破れ Weinberg - Salam Model

$H_{eff} = \frac{1}{M} l_L l_L \varphi \varphi \longrightarrow \langle \varphi \rangle = \begin{pmatrix} 0 \\ v \end{pmatrix} / \sqrt{2}$

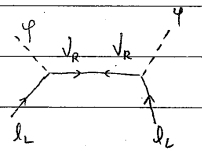
$m_\nu = \frac{v^2}{2M} \simeq 1\,eV$

$\nu_{eff} = \left(\frac{v^2}{2M}\right) \nu_L \nu_L$



B-L ≠ 0 −1, −1, 0, 0

$\mathcal{L} = M \nu_R \nu_R$



$\sim \frac{1}{M_R} \varphi\varphi\, l_L l_L$

SO(10) $16 = 5^* + 10 + 1 = \left(\begin{pmatrix} \nu \\ e \end{pmatrix}_L \begin{pmatrix} \nu \\ e \end{pmatrix}_R \begin{pmatrix} u \\ d \end{pmatrix}_L \begin{pmatrix} u \\ d \end{pmatrix}_R \right)$

→ SU(3) × SU(2) × U(1)

レポート 9/1 まで